

Sets, Functions, and Cardinality

COMP2711 Discrete Mathematical Tools

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SECTION 1

Sets

Overview

Every student in a class needs a locker.

1. Which **students and lockers** are under consideration?
2. What **assignment** sends each student to a locker?
3. Can it avoid sharing and use every locker?

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sets $\longrightarrow$ **functions** $\longrightarrow$ **cardinality**

Sets

A **set** is an unordered collection of distinct objects.

$x \in A$ means “ x is an element of A .”

Throughout this course,

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ natural numbers (including 0)

$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ integers

$\mathbb{R}$ real numbers: all points on the number line

Order and repetition do not create new elements.

Describing sets

Let O be the odd natural numbers below 10.

List all elements: $O = \{1, 3, 5, 7, 9\}$

Set builder $O = \{n \in \mathbb{N} \mid n < 10 \text{ and } n \text{ is odd}\}$

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A set builder needs a domain. It reads:

“the elements n in $\mathbb{N}$ for which the condition is true.”

Examples of sets

Sets can contain many kinds of objects.

Strings

$S = \{\text{"Ada"}, \text{"Ben"}, \text{"Chen"}\}$

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Mixed objects	$M = \{7, \text{"hello"}, \{1, 2\}\}$
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In M , the number 7, the string "hello", and the set $\{1, 2\}$ are three separate elements.

Set equality

MUST UNDERSTAND A standard set is unordered and deduplicated.

Unordered

Order does not matter:

$$\{1, 3, 5\} = \{5, 3, 1\}.$$

Deduplicated

Repetition is ignored:

$$\{1, 3, 3, 5\} = \{1, 3, 5\}.$$

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$$A = B \iff \forall x (x \in A \leftrightarrow x \in B).$$

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Equivalently,

$$A = B \iff A \subseteq B \text{ and } B \subseteq A.$$

Membership and subsets

MUST UNDERSTAND $\in$ compares one object with a set; $\subseteq$ compares two sets.

Membership

$$x \in A$$

The object x is an element of A .

Subset

$$B \subseteq A \iff \forall x (x \in B \rightarrow x \in A)$$

Every element of B is also an element of A .

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Proper subset

$$B \subsetneq A \iff B \subseteq A \text{ and } B \neq A.$$

Thus, A contains at least one element not in B .

Membership and subset examples

Let $A = \{1, \{2, 3\}\}$.

Statement	True?	Reason
$1 \in A$		
$\{1\} \subseteq A$		
$\{2, 3\} \in A$		
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$\{2, 3\} \in A$	yes	that set is one element of A
$2 \in A$	no	being inside an element is not membership

The braces matter: 1 is a number, while $\{1\}$ is a set.

Set size

MUST UNDERSTAND For a finite set S , its size is the number of distinct elements.

The size of S , also called its **cardinality**, is written $|S|$.

Examples

Set	Size
$S = \{a, b, c\}$	$ S = 3$
$T = \{1, 1, 2\} = \{1, 2\}$	$ T = 2$

Repeating an element does not increase the size of a set.

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The empty set

$$\emptyset = \{ \}$$

It contains no elements, so

$$|\emptyset| = 0.$$

It is the **only set** of size 0.

Repeating an element does not increase the size of a set.

Power sets

$$\mathcal{P}(S) = \{T \mid T \subseteq S\}$$

For $S = \{a, b\}$, every subset appears in the complete list:

Choice bits	00	01	10	11
Subset	$\emptyset$	$\{b\}$	$\{a\}$	$\{a, b\}$

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Two independent yes/no choices give $|\mathcal{P}(S)| = 2^2 = 4$. In general, $|S| = n$ gives $|\mathcal{P}(S)| = 2^n$.

Cartesian products

The **Cartesian product** is

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

For $A = \{1, 2\}$ and $B = \{x, y, z\}$,

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

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Each element is an **ordered pair**: the first and second positions have different roles.

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

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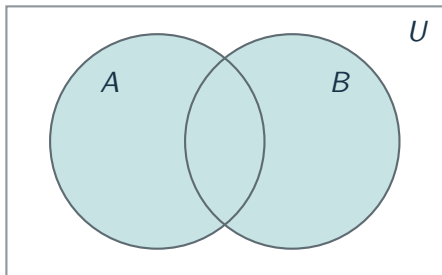
$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

If A and B are finite, $|A \times B| = |A||B|$.

SECTION 2

Set operations

Union

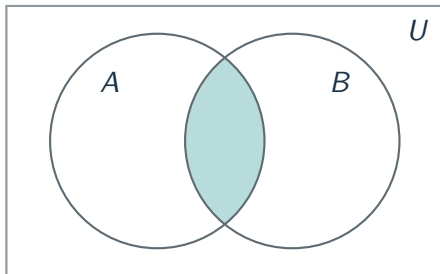


$$A \cup B$$

Shade every point that belongs to **at least one** of the sets.

$$x \in A \cup B \iff x \in A \vee x \in B.$$

Intersection



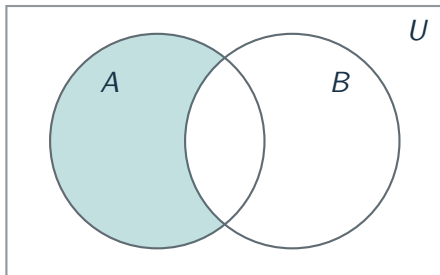
$$A \cap B$$

Shade only the points that belong to **both** sets.

$$x \in A \cap B \iff x \in A \wedge x \in B.$$

A and B are **disjoint** when $A \cap B = \emptyset$.

Difference



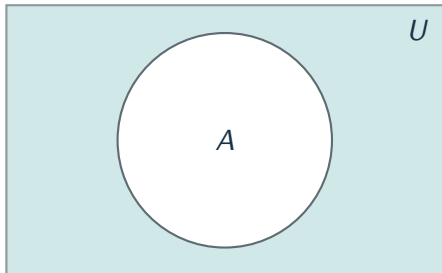
$$A \setminus B$$

Start with A , then remove every point that also belongs to B .

$$x \in A \setminus B \iff x \in A \wedge x \notin B.$$

In general, $A \setminus B \neq B \setminus A$.

Complement



$\bar{A}$

Shade everything in the universal set U that lies **outside** A .

$$x \in \bar{A} \iff x \in U \wedge x \notin A.$$

The complement depends on the chosen universe U .

Set identities and logic

Membership translates set operations into connectives:

$$\cap \longleftrightarrow \wedge, \quad \cup \longleftrightarrow \vee, \quad x \in \bar{A} \longleftrightarrow \neg(x \in A).$$

So De Morgan's logical law gives

$$\neg(P \vee Q) \equiv \neg P \wedge \neg Q \implies \overline{A \cup B} = \bar{A} \cap \bar{B}.$$

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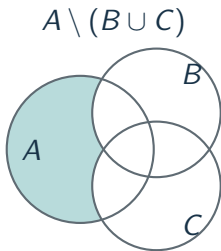
When an identity is uncertain, follow an arbitrary element.

Worked example: Venn diagram

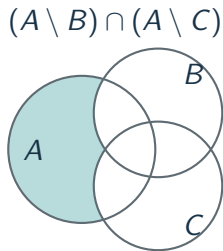
A **Venn diagram** draws sets as regions.

Compare the two sides of

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C).$$



Remove everything in B or C from A .



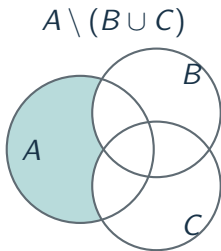
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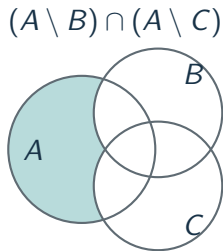
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Remove everything in B or C from A .

Both sides shade the points in A that are in neither B nor C .



Keep points in both differences.

Worked example: element proof

Prove $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

$$x \in A \setminus (B \cup C)$$

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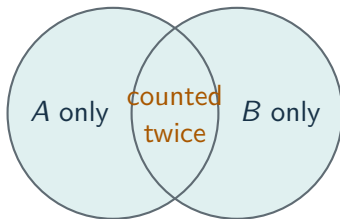
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The membership conditions agree for every x , so the sets are equal.

Inclusion–exclusion

Adding $|A| + |B|$ counts the intersection twice.



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Families of sets

For sets $\{A_i\}_{i \in I}$,

$$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I (x \in A_i)\},$$

$$\bigcap_{i \in I} A_i = \{x \mid \forall i \in I (x \in A_i)\}.$$

Union asks for **at least one**; intersection asks for **every**.

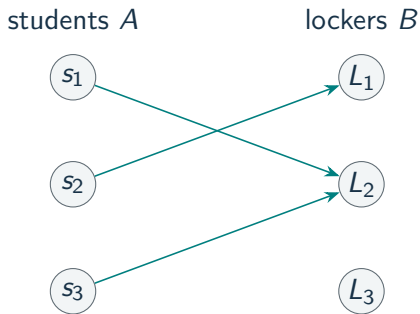
SECTION 3

Functions

Functions

MUST UNDERSTAND Every input has exactly one output.

A **function** $f : A \rightarrow B$ assigns exactly one element of B to each element of A .



Two students may share a locker. That is still a function.

Codomain and range

For a function $f : A \rightarrow B$:

Domain	A
Codomain	B
Range	elements in B <i>touched</i> by f : $f[A]$

$$f[A] = \{f(a) \mid a \in A\} \subseteq B.$$

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Example. $f : \{1, 2, 3\} \rightarrow \{a, b, c\}$, where $f(1) = a$, $f(2) = a$, and $f(3) = c$.

Range = $\{a, c\}$. b is in the codomain, but not the range.

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Changing only the codomain can change whether a function is onto.

Recognizing functions

Each rule starts with students $\{s_1, s_2, s_3\}$.

Rule	Assignment	Function?
A	$s_1 \mapsto L_1, s_2 \mapsto L_1, s_3 \mapsto L_2$	
B	$s_1 \mapsto L_1$ and $s_1 \mapsto L_2; s_2 \mapsto L_1, s_3 \mapsto L_2$	
C	$s_1 \mapsto L_1, s_2 \mapsto L_2; s_3$ unassigned	

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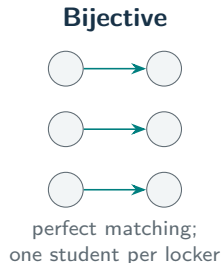
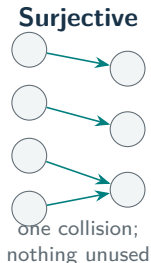
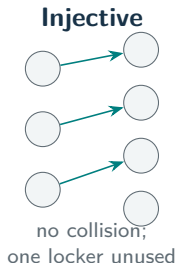
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A function may reuse outputs. It may not split or omit an input.

Injective, surjective, and bijective functions

MUST UNDERSTAND Separate collisions from unused targets.



Definitions using quantifiers

Injective:

$$\forall a_1, a_2 \in A, \quad f(a_1) = f(a_2) \rightarrow a_1 = a_2,$$

$$\text{Contrapositive: } \forall a_1, a_2 \in A, \quad a_1 \neq a_2 \rightarrow f(a_1) \neq f(a_2).$$

Equal outputs force equal inputs; distinct inputs have distinct outputs.

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Every target has at least one preimage.

Bijjective:

$$f \text{ is bijective} \iff f \text{ is injective and surjective.}$$

Every target is reached by exactly one input.

Example: squaring on the integers

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It is **not surjective**: -1 has no integer preimage.

Its range is $\{0, 1, 4, 9, \dots\}$, not the full codomain $\mathbb{Z}$.

Changing the domain and codomain

Restrict both sides:

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If $g(m) = g(n)$, then $m^2 = n^2$; since $m, n \geq 0$, we get $m = n$.

Every target is $k^2 = g(k)$ for some $k \in \mathbb{N}$.

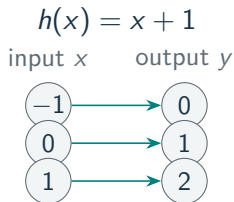
So g is **bijective**. The formula stayed the same; the sets changed.

Inverse functions

A function $f : A \rightarrow B$ has an inverse $f^{-1} : B \rightarrow A$ exactly when f is bijective.

It reverses each pair: $f^{-1}(b) = a \iff f(a) = b$.

Example. Let $h : \mathbb{Z} \rightarrow \mathbb{Z}$ be $h(x) = x + 1$.



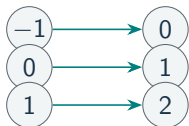
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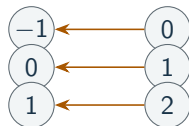
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$h(x) = x + 1$
input x output y



$h^{-1}(y) = y - 1$
output x input y



Composition

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Let $g(x) = x + 1$ and $f(x) = 2x$ on $\mathbb{Z}$.

$$(f \circ g)(x) = 2(x + 1) = 2x + 2.$$

Apply g first, then apply f .

Composition laws

Associative. Whenever the compositions are defined,

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Generally not commutative. Let $g(x) = x + 1$ and $f(x) = 2x$ on $\mathbb{Z}$.

$$(f \circ g)(x) = 2x + 2, \quad (g \circ f)(x) = 2x + 1.$$

Therefore $f \circ g \neq g \circ f$. Changing the order can change the function.

Floor and ceiling

$\lfloor x \rfloor$ = greatest integer at most x , $\lceil x \rceil$ = least integer at least x .



Floor and ceiling

$\lfloor x \rfloor$ = greatest integer at most x , $\lceil x \rceil$ = least integer at least x .



$$-3 \leq -2.7 < -2, \quad \lfloor -2.7 \rfloor = -3, \quad \lceil -2.7 \rceil = -2.$$

Counting functions

Let $|A| = n$ and $|B| = m$.

Each input has m independent choices, so

$$\#\{f : A \rightarrow B\} = m^n.$$

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Each input has m independent choices, so

$$\#\{f : A \rightarrow B\} = m^n.$$

For an injection, outputs cannot be reused:

$$\#\{\text{injections } A \rightarrow B\} = \begin{cases} m(m-1) \cdots (m-n+1), & m \geq n, \\ 0, & m < n. \end{cases}$$

Between two n -element sets: n^n functions and $n!$ bijections.

Set practice

Let

$$A = \{1, 2\}, \quad B = \{2, 3\}.$$

Compute:

1. $A \cup B$
2. $A \cap B$
3. $A \setminus B$
4. $\mathcal{P}(A)$
5. $A \times B$

Write every resulting set by listing all of its elements.

Set practice solution

For $A = \{1, 2\}$ and $B = \{2, 3\}$:

$$A \cup B = \{1, 2, 3\},$$

$$A \cap B = \{2\},$$

$$A \setminus B = \{1\},$$

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\},$$

$$A \times B = \{(1, 2), (1, 3), (2, 2), (2, 3)\}.$$

Function practice

Let $f, g : \mathbb{Z} \rightarrow \mathbb{Z}$ be

$$f(n) = n + 3, \quad g(n) = 2n.$$

1. Find $(f \circ g)(n)$ and $(g \circ f)(n)$.
2. Decide whether each function is injective, surjective, or bijective.
3. If a function is bijective, find its inverse.

Remember that both the domain and codomain are $\mathbb{Z}$.

Function practice solution

$$(f \circ g)(n) = f(2n) = 2n + 3,$$
$$(g \circ f)(n) = g(n + 3) = 2n + 6.$$

Function	Classification and reason
f	Bijjective: adding 3 gives a distinct output for every input and reaches every integer.
g	Injective but not surjective: its outputs are even, so it misses every odd integer.

Since f is bijective,

$$f^{-1}(n) = n - 3.$$

SECTION 4

Cardinality

Hilbert's hotel

How An Infinite Hotel Ran Out Of Room

<https://youtu.be/OxGsU8oIWjY?si=CJF3Fb0yVXWF0mlG>

Highly recommended

Math's Fundamental Flaw

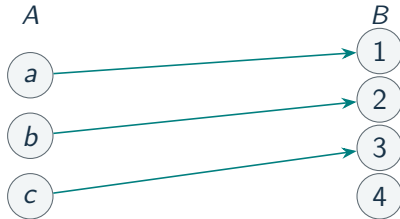
How paradoxes, infinity, and formal logic led to modern computation

<https://www.youtube.com/watch?v=HeQX2HjkcNo>

Finite cardinality

For a finite set, count its distinct elements.

$$A = \{a, b, c\}, \quad B = \{1, 2, 3, 4\}.$$



$$|A| = 3 < 4 = |B|.$$

The unmatched element confirms that the finite set B is larger.

Infinite cardinality

How can we compare the sizes of two infinite sets?

Can one infinity be larger than another?

Infinite cardinality

Yes.

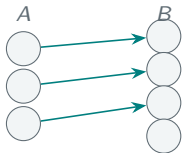
Georg Cantor compared infinite sets by **matching their elements**.

Infinite cardinality

MUST UNDERSTAND The most important concept in this section.

Injection $A \rightarrow B$

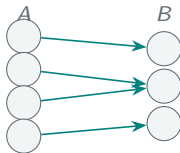
$$|A| \leq |B|$$



No collisions.

Surjection $A \rightarrow B$

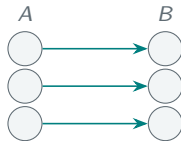
$$|A| \geq |B|$$



Every element of B is reached.

Bijection $A \rightarrow B$

$$|A| = |B|$$



No collisions, and every element of B is reached.

Schroeder–Bernstein theorem

If

$$|A| \leq |B| \quad \text{and} \quad |B| \leq |A|,$$

then

$$\boxed{|A| = |B|}.$$

Schroeder–Bernstein theorem

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Equivalent statement

If there is an injection

$$f : A \rightarrow B$$

and an injection

$$g : B \rightarrow A,$$

then there is a bijection $h : A \rightarrow B$.

Schroeder–Bernstein theorem

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Equivalent statement

If there is an injection

$$f : A \rightarrow B$$

and an injection

$$g : B \rightarrow A,$$

then there is a bijection $h : A \rightarrow B$.

Proof: non-trivial and outside the scope of this lecture.

Example 1: Natural numbers and even numbers

$$\text{Even} = \{0, 2, 4, 6, \dots\} \subsetneq \mathbb{N}.$$

The inclusion map

$$i : \text{Even} \rightarrow \mathbb{N}, \quad i(x) = x$$

is injective: distinct even numbers remain distinct.

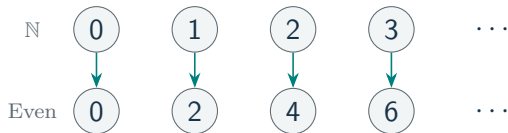
Therefore $|\text{Even}| \leq |\mathbb{N}|$.

Example 1: Natural numbers and even numbers

$$\text{Even} = \{0, 2, 4, 6, \dots\} \subsetneq \mathbb{N}.$$

But there is also an injection in the other direction:

$$g : \mathbb{N} \rightarrow \text{Even}, \quad g(x) = 2x.$$



The surprise: $\mathbb{N}$ injects into its proper subset Even , so $|\mathbb{N}| \leq |\text{Even}|$.

Example 1: Natural numbers and even numbers

$$\text{Even} = \{0, 2, 4, 6, \dots\} \subsetneq \mathbb{N}.$$

Is $g : \mathbb{N} \rightarrow \text{Even}$, where $g(x) = 2x$, actually a bijection?

Prove that it is both injective and surjective.

Example 1: Natural numbers and even numbers

$$\text{Even} = \{0, 2, 4, 6, \dots\} \subsetneq \mathbb{N}.$$

Injective, using the contrapositive definition:

$$x_1 \neq x_2 \implies 2x_1 \neq 2x_2.$$

Distinct inputs have distinct outputs, so there is no collision.

Surjective onto Even: every $y \in \text{Even}$ has the form $y = 2x$ for some $x \in \mathbb{N}$.

$$\boxed{|\mathbb{N}| = |\text{Even}|}$$

A proper subset of an infinite set can have the same cardinality as the full set.

Example 2: An interval and the real line

$$\text{Is } |(0, 1)| = |\mathbb{R}|?$$



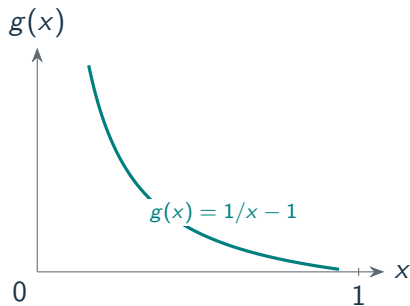
Example 2: An interval and the real line

$$\text{Is } |(0, 1)| = |\mathbb{R}|?$$



Yes. Let's find a bijection between them.

Example 2, Step 1: $(0, 1)$ and $(0, \infty)$



Find a bijection from $(0, 1)$ to the positive real numbers:

$$g : (0, 1) \rightarrow (0, \infty), \quad g(x) = \frac{1}{x} - 1.$$

$$x \rightarrow 0^+ \Rightarrow g(x) \rightarrow \infty,$$

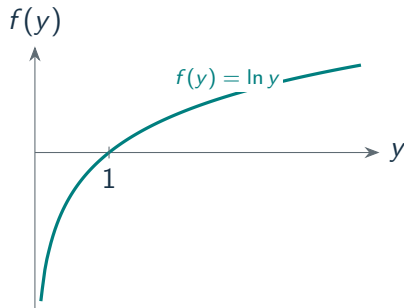
$$x \rightarrow 1^- \Rightarrow g(x) \rightarrow 0^+.$$

The inverse is

$$g^{-1}(y) = \frac{1}{y+1}.$$

$| (0, 1) | = | (0, \infty) |$

Example 2, Step 2: $(0, \infty)$ and $\mathbb{R}$



The logarithm is a bijection from the positive real numbers to all real numbers:

$$f : (0, \infty) \rightarrow \mathbb{R}, \quad f(y) = \ln y.$$

$$y \rightarrow 0^+ \Rightarrow f(y) \rightarrow -\infty,$$

$$y \rightarrow \infty \Rightarrow f(y) \rightarrow \infty.$$

The inverse is

$$f^{-1}(z) = e^z.$$

$$\boxed{|(0, \infty)| = |\mathbb{R}|}$$

Example 2, Step 3: Composition

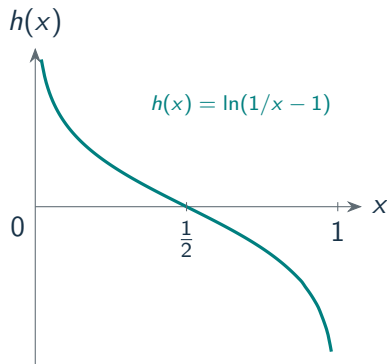


The composition maps directly from $(0, 1)$ to $\mathbb{R}$:

$$h = f \circ g : (0, 1) \rightarrow \mathbb{R},$$

$$h(x) = f(g(x)) = \ln\left(\frac{1}{x} - 1\right).$$

Example 2, Step 3: The composed function



The graph covers every real output:

$$x \rightarrow 0^+ \Rightarrow h(x) \rightarrow +\infty,$$

$$x \rightarrow 1^- \Rightarrow h(x) \rightarrow -\infty.$$

Its inverse is

$$h^{-1}(z) = \frac{1}{1 + e^z}.$$

A composition of bijections is a bijection.
Therefore,

$$\boxed{|(0, 1)| = |\mathbb{R}|}.$$

Must Understand: Countable sets

1. Intuitive meaning

A set S is **countable** if we can list all its elements:

$$s_1, s_2, s_3, \dots$$

Every element of S appears somewhere in the list. A finite list may stop.

2. Formal definition

S is countable if it is finite, or if there is a bijection

$$e : \mathbb{N} \longrightarrow S.$$

Equivalently,

$$\boxed{|S| \leq |\mathbb{N}|}.$$

Countable case 1: Integers

Theorem. The integers $\mathbb{Z}$ are countable.

Proof. To prove $|\mathbb{Z}| \leq |\mathbb{N}|$, we just need to find a surjection

$$e : \mathbb{N} \longrightarrow \mathbb{Z}.$$

Intuitively, this is the same as listing all integers.

Countable case 1: Integers

Let $e : \mathbb{N} \rightarrow \mathbb{Z}$ be defined by

$$e(0) = 0, \quad e(2k - 1) = k, \quad e(2k) = -k \quad (k \geq 1).$$

This gives the list

$$0, 1, -1, 2, -2, 3, -3, \dots$$

Exercise. Show that e is bijective.

Countable case 2: Rational numbers

Theorem. $\mathbb{Q}$ is countable.

$$\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}.$$

Proof. To prove $|\mathbb{Q}| \leq |\mathbb{N}|$, we just need to find a surjection

$$e : \mathbb{N} \longrightarrow \mathbb{Q}.$$

In other words, list the rational numbers so that every rational appears.

Countable case 2: Rational numbers

Every positive rational is p/q for positive integers p, q .

$p + q$	Pairs (p, q)	New values p/q
2	(1, 1)	1
3	(1, 2), (2, 1)	$1/2, 2$
4	(1, 3), (2, 2), (3, 1)	$1/3, 1$ (repeat), 3
5	(1, 4), (2, 3), (3, 2), (4, 1)	$1/4, 2/3, 3/2, 4$

Countable case 2: Rational numbers

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5	(1, 4), (2, 3), (3, 2), (4, 1)	1/4, 2/3, 3/2, 4

Continue by constant $p + q$ and skip any value already listed. Every positive rational eventually appears.

Countable case 2: Rational numbers

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5	(1, 4), (2, 3), (3, 2), (4, 1)	$1/4, 2/3, 3/2, 4$

Continue by constant $p + q$ and skip any value already listed. Every positive rational eventually appears.

If $q_1, q_2, \dots$ is this positive list, then

$$0, q_1, -q_1, q_2, -q_2, \dots$$

lists every rational exactly once.

Countable case 3: Cartesian products

Theorem.

If A and B are countable, then their Cartesian product

$A \times B$ is countable.

Countable case 3: Proof

Proof. Suppose A and B are countably infinite:

$$A = \{a_0, a_1, \dots\}, \quad B = \{b_0, b_1, \dots\}$$

List pairs by increasing $i + j$:

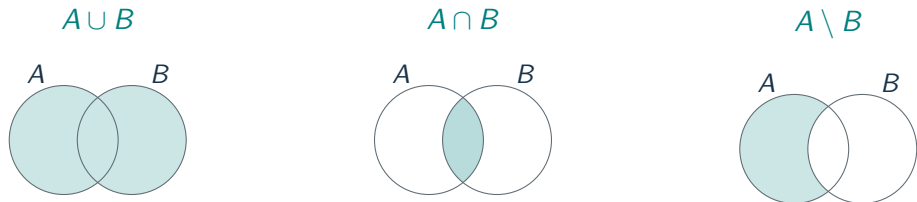
$i + j$	pairs
0	(a_0, b_0)
1	$(a_0, b_1), (a_1, b_0)$
2	$(a_0, b_2), (a_1, b_1), (a_2, b_0)$
$\vdots$	$\vdots$

Every diagonal is finite, and every pair appears on one diagonal.

If either set is finite, the same listing argument is simpler.

Countable case 4: Set operations

Theorem. If A and B are countable, then all three sets below are countable.



Exercise: prove all three statements. A complete proof is in the course note.

Must Understand: Strict cardinality and uncountable sets

Strict cardinality

If an injection $A \rightarrow B$ exists but no bijection exists, we write

$$|A| < |B|.$$

A is strictly smaller than B in cardinality.

Uncountable sets

A set T is **uncountable** if

$$|T| > |\mathbb{N}|.$$

Intuitively, there is no list

$$t_1, t_2, t_3, \dots$$

that contains every element of T .

Uncountable case 1: Real numbers

We already proved

$$|(0, 1)| = |\mathbb{R}|.$$

Therefore it is enough to show that $(0, 1)$ is uncountable.

Suppose, for contradiction, that every number in $(0, 1)$ appears in a list

$$r_1, r_2, r_3, \dots$$

Cantor's idea: construct a number in $(0, 1)$ that is missing from this list.

Cantor's diagonal argument

Suppose $r_i = 0.d_{i1}d_{i2}d_{i3}\dots$ lists every number in $(0, 1)$. Choose expansions that do not end in infinitely many 9s.

For a concrete proposed list, focus on the diagonal:

$$r_1 = 0.\textcolor{teal}{2}379\dots$$

$$r_2 = 0.4\textcolor{teal}{4}59\dots$$

$$r_3 = 0.09\textcolor{teal}{1}8\dots$$

$$r_4 = 0.805\textcolor{teal}{5}\dots$$

Cantor's diagonal argument

Suppose $r_i = 0.d_{i1}d_{i2}d_{i3}\dots$ lists every number in $(0, 1)$. Choose expansions that do not end in infinitely many 9s.

For a concrete proposed list, focus on the diagonal:

$$r_1 = 0.2379\dots$$

$$r_2 = 0.4459\dots$$

$$r_3 = 0.0918\dots$$

$$r_4 = 0.8055\dots$$

Choose

$$c_i = \begin{cases} 4, & d_{ii} \neq 4, \\ 5, & d_{ii} = 4, \end{cases} \quad r = 0.c_1c_2c_3c_4\dots$$

$(0, 1)$ is uncountable

The constructed r differs from r_i at decimal position i .

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Therefore

$$r \neq r_1, \quad r \neq r_2, \quad r \neq r_3, \quad \dots$$

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Therefore

$$r \neq r_1, \quad r \neq r_2, \quad r \neq r_3, \quad \dots$$

r lies in $(0, 1)$, but it is absent from the supposed complete list.

$(0, 1)$ is uncountable

The constructed r differs from r_i at decimal position i .

Therefore

$$r \neq r_1, \quad r \neq r_2, \quad r \neq r_3, \quad \dots$$

r lies in $(0, 1)$, but it is absent from the supposed complete list.

Contradiction. The interval $(0, 1)$, and hence $\mathbb{R}$, is **uncountable**.

$(0, 1)$ is uncountable

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Therefore

$$r \neq r_1, \quad r \neq r_2, \quad r \neq r_3, \quad \dots$$

r lies in $(0, 1)$, but it is absent from the supposed complete list.

Contradiction. The interval $(0, 1)$, and hence $\mathbb{R}$, is **uncountable**.

This is **Cantor diagonalization**.

Uncountable case 2: Cantor's theorem

$$|S| < |\mathcal{P}(S)| \quad \text{for every set } S.$$

That is, the power set of S is “bigger” than S in cardinality.

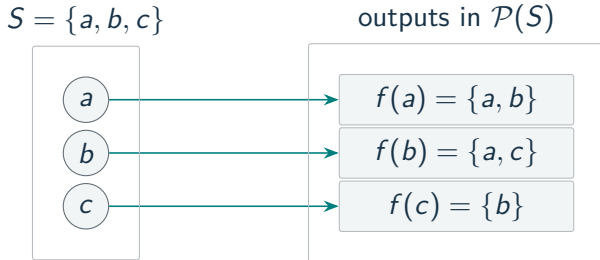
Consequence

If S is countably infinite, then $\mathcal{P}(S)$ is uncountable.

Proof. The map $x \mapsto \{x\}$ is injective, so $|S| \leq |\mathcal{P}(S)|$. This direction is left as an exercise.

It remains to prove that every function $f : S \rightarrow \mathcal{P}(S)$ is **not surjective**. Hence **no bijection $S \rightarrow \mathcal{P}(S)$ exists**.

A function into a power set



This is one possible f : each element of S maps to **one subset of S** .

We want to prove that **every** function $f : S \rightarrow \mathcal{P}(S)$ misses at least one subset of S .
Therefore no such function is surjective.

Constructing the diagonal subset

For any $f : S \rightarrow \mathcal{P}(S)$, define

$$D = \{x \in S \mid x \notin f(x)\} \in \mathcal{P}(S).$$

It collects the elements that do not belong to their own assigned subsets.

Input	Assigned subset	Self-membership	Decision for D
a	$f(a) = \{a, b\}$		
b	$f(b) = \{a, c\}$		
c	$f(c) = \{b\}$		

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Input	Assigned subset	Self-membership	Decision for D
a	$f(a) = \{a, b\}$	$a \in f(a)$: yes	leave a out
b	$f(b) = \{a, c\}$		
c	$f(c) = \{b\}$		

Constructing the diagonal subset

For any $f : S \rightarrow \mathcal{P}(S)$, define

$$D = \{x \in S \mid x \notin f(x)\} \in \mathcal{P}(S).$$

It collects the elements that do not belong to their own assigned subsets.

Input	Assigned subset	Self-membership	Decision for D
a	$f(a) = \{a, b\}$	$a \in f(a)$: yes	leave a out
b	$f(b) = \{a, c\}$	$b \in f(b)$: no	put b in
c	$f(c) = \{b\}$		

Constructing the diagonal subset

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c	$f(c) = \{b\}$	$c \in f(c)$: no	put c in

For this example, $D = \{b, c\}$; it is not one of the three outputs.

Why D is not reached

$$D = \{x \in S \mid x \notin f(x)\}.$$

D collects all items not contained in their corresponding subsets.

Goal: show that no $y \in S$ satisfies $f(y) = D$. Then f misses D , so f is not surjective.

Proof by contradiction. Assume some $y \in S$ satisfies $f(y) = D$.

Why D is not reached

Recall the definition of D and our contradiction assumption:

$$D = \{x \in S \mid x \notin f(x)\}, \quad f(y) = D.$$

If $y \in D$:

Every $x \in D$ satisfies $x \notin f(x)$.

Therefore $y \notin f(y) = D$, so $y \notin D$.

Contradiction.

If $y \notin D$:

Since $f(y) = D$ and $y \notin D$, we have $y \notin f(y)$. The definition of D therefore collects y , so $y \in D$. Contradiction.

Thus no such y exists. Therefore D is not reached by f , and f is not surjective.

Uncountable case 3: Irrational numbers

The real numbers split into rational and irrational numbers:

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q}).$$

We already know

$\mathbb{Q}$ is countable, $\mathbb{R}$ is uncountable.

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Suppose the irrational numbers $\mathbb{R} \setminus \mathbb{Q}$ were countable. Then the union of two countable sets would make $\mathbb{R}$ countable.

This is a contradiction. Therefore

$\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

Reals and power sets

Recall: $\mathbb{N} = \{0, 1, 2, \dots\}$ is the set of natural numbers.

Theorem. The power set of $\mathbb{N}$ has the same cardinality as the real numbers:

$$|\mathcal{P}(\mathbb{N})| = |\mathbb{R}|.$$

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Hint: choice bits

A subset $A \subseteq \mathbb{N}$ makes one choice for each natural number:

$$b_n = \begin{cases} 1, & n \in A, \\ 0, & n \notin A. \end{cases} \quad \longleftrightarrow \quad 0.b_0b_1b_2\dots \text{ in binary.}$$

Infinite cardinal notation

Every countably infinite set S has the same cardinality as $\mathbb{N}$:

$$|S| = |\mathbb{N}| = \aleph_0.$$

The symbol $\aleph_0$ (“aleph null”) denotes the **smallest infinite cardinality**.

The symbol $\aleph_1$ (“aleph one”) denotes the **smallest uncountable cardinality**.

$$\aleph_0 < \aleph_1.$$

The continuum hypothesis

Define the cardinality of the real numbers as **the continuum**:

$$\mathfrak{c} := |\mathbb{R}|.$$

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$$\mathfrak{c} := |\mathbb{R}|.$$

We already know $\mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})|$. By Cantor's theorem,

$$\mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| > |\mathbb{N}| = \aleph_0.$$

The continuum hypothesis

Define the cardinality of the real numbers as **the continuum**:

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$$\mathfrak{c} = |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| > |\mathbb{N}| = \aleph_0.$$

The **continuum hypothesis** asks a simple yet profound question:

Is the cardinality of the real numbers the **smallest uncountable cardinality**?

$$\mathfrak{c} = \aleph_1 ?$$

Equivalently: is there no cardinality strictly between the sizes of $\mathbb{N}$ and $\mathbb{R}$?

Optional: CH is independent of ZFC

Answer: CH is **independent of ZFC**.

ZFC is the standard axiom system used to formalize most modern mathematics.

“Independent” means that, using the axioms of ZFC, we can prove neither CH nor $\neg$ CH. In other words, ZFC cannot decide whether CH is true or false.

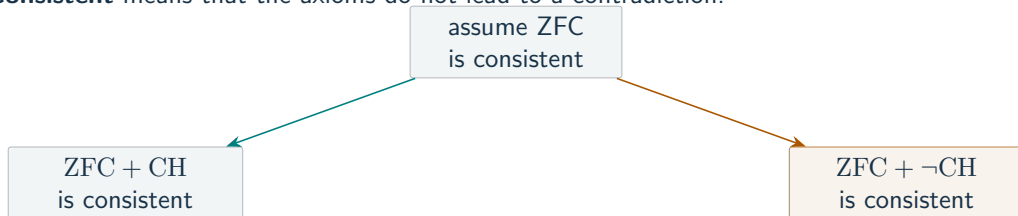
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“Independent” means that, using the axioms of ZFC, we can prove neither CH nor $\neg$ CH. In other words, ZFC cannot decide whether CH is true or false.

Consistent means that the axioms do not lead to a contradiction.



Assuming ZFC is consistent, adding either CH or $\neg$ CH still gives a consistent axiom system.

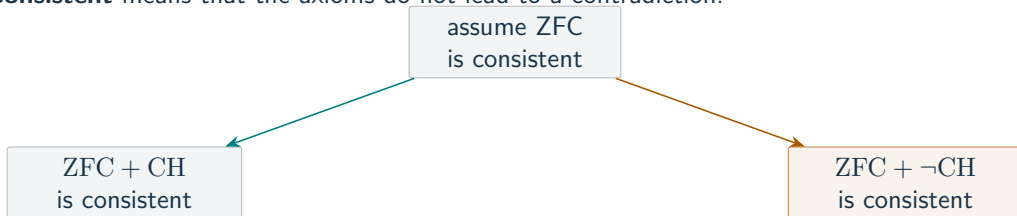
Optional: CH is independent of ZFC

Answer: CH is **independent of ZFC**.

ZFC is the standard axiom system used to formalize most modern mathematics.

“Independent” means that, using the axioms of ZFC, we can prove neither CH nor $\neg$ CH. In other words, ZFC cannot decide whether CH is true or false.

Consistent means that the axioms do not lead to a contradiction.



Assuming ZFC is consistent, adding either CH or $\neg$ CH still gives a consistent axiom system.

Gödel (1940) ZFC cannot disprove CH, assuming ZFC is consistent.

Cohen (1963) ZFC cannot prove CH, assuming ZFC is consistent.

SECTION 5

Worked problems

Worked problem: classifying a function

Let

$$f : \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(n) = 3n - 2.$$

Decide whether f is injective and surjective. If an inverse exists, find it.

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Plan:

1. For injection, start from equal outputs.
2. For surjection, start from an arbitrary target $y \in \mathbb{Z}$.

Solution

Injection. Suppose $f(a) = f(b)$.

$$3a - 2 = 3b - 2 \implies a = b.$$

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For $y = 0$, this is not an integer. So f is not surjective.

No inverse $\mathbb{Z} \rightarrow \mathbb{Z}$ exists. Restrict the codomain to $\{3n - 2 \mid n \in \mathbb{Z}\}$ to obtain a bijection.

Summary

1. Sets are determined by membership; set algebra is logic applied to membership.
2. Functions transform elements. Injection, surjection, and bijection answer different matching questions.
3. Cardinality compares sizes by functions. Diagonalization proves $\aleph_0 < \mathfrak{c}$; CH asks whether $\mathfrak{c} = \aleph_1$.

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1. Sets are determined by membership; set algebra is logic applied to membership.
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Next: **mathematical induction** proves a claim for every natural number by linking one case to the next.